SmartSupport AI

Risk Assessment & Mitigation Plan
Stakeholder Analysis

Project 5: Agentic AI Customer Support Platform

1. Project Overview

SmartSupport AI is a customer-support platform that uses a Large Language Model (LLM) to answer customer questions, retrieve relevant help articles, perform selected support actions, and transfer complex cases to human agents. The proposed system includes a customer chat widget, conversation and message storage, a knowledge base, support tickets, agent and admin dashboards, analytics, and audit logs.

The proposed technology stack includes ASP.NET Core MVC, Entity Framework Core, ASP.NET Identity, OpenAI API or Azure OpenAI, Semantic Kernel, SignalR, Azure Cognitive Search, Redis, Azure Blob Storage, Chart.js, Bootstrap 5, and Microsoft Azure.

This document focuses on the analysis and planning side of the project. Risk ratings are initial estimates based on the project proposal and should be reviewed by the team as implementation details become clearer.

2. Stakeholder Analysis

Stakeholders are people or groups who use the system, operate it, fund it, maintain it, or may be affected by its behavior. Understanding their needs helps the team prioritize requirements and manage risks.

Stakeholder	Role / Interest	Influence	Impact on Project	Main Needs / Expectations
Customers / End Users	Ask questions, get accurate answers, check order information, and receive help quickly.	High	High	Simple chat experience; accurate, understandable answers; privacy; clear handoff to a human.
Human Support Agents	Handle escalated conversations, manage tickets, review AI summaries, and help customers.	High	High	Useful summaries, clear ticket priorities, access to conversation history, ability to correct AI suggestions.
Support Team Lead / Manager	Monitor service quality, workload, escalation rates, and customer satisfaction.	High	Medium–High	Reliable analytics, queue visibility, performance indicators, and consistent workflows.
System Administrator	Configure users, roles, prompts, knowledge articles, integrations, and system settings.	High	Medium	Secure admin access, manageable configuration, logs, and operational controls.
Project Team / Developers	Design, implement, test, document, and deploy the application.	High	High	Clear scope, stable architecture, test data, API documentation, and realistic milestones.
Business / Product Owner	Defines support goals, approves scope, and evaluates whether the system meets business needs.	High	Medium–High	Reduced response time, acceptable operating cost, measurable service quality, and controlled risk.
AI / Cloud Service Providers	Provide the external LLM and cloud/search services used by the platform.	Medium	Medium	Valid API configuration, service availability, usage limits, and compliance with provider terms.
Security / Privacy Responsible Person	Reviews access controls, sensitive data handling, retention, and auditability.	High	Medium	Least-privilege access, appropriate data retention, protection of personal information, and traceable actions.

2.1 Stakeholder Engagement Plan

Stakeholder Group	Engagement Approach
Customers	Collect feedback through short usability tests and a post-chat rating; review confusing or unanswered questions.
Support agents	Involve agents in defining escalation rules, ticket statuses, and the information shown in the agent dashboard.
Managers / Product owner	Agree on success measures and review a brief progress report at milestones.
Administrators	Validate roles, configuration screens, access permissions, and audit-log needs.
Development team	Use regular task reviews, code reviews, and integration checks; record decisions and unresolved issues.
Security / privacy reviewer	Review data flows, permissions, API-key handling, logging, and retention before deployment.

3. Risk Assessment Method

Each risk is scored using two simple scales: Likelihood and Impact, each from 1 to 5. Risk Score = Likelihood × Impact. The score is used to prioritize mitigation; it is not a statistical prediction.

Score	Likelihood	Impact
1	Rare	Negligible effect on scope or users
2	Unlikely	Minor inconvenience or limited rework
3	Possible	Noticeable disruption or moderate rework
4	Likely	Major feature, schedule, cost, or user impact
5	Almost certain	Severe disruption, safety, privacy, or delivery impact

Priority guide: 1–4 = Low; 5–9 = Medium; 10–16 = High; 17–25 = Critical. The team should reassess scores when controls are implemented or project assumptions change.

4. Risk Register and Mitigation Plan

ID	Risk	L	I	Score / Priority	Mitigation / Response	Owner	Residual Risk*
R1	AI generates inaccurate or misleading answers	4	5	20 — Critical	Use approved knowledge sources for factual answers; instruct the model to say when it cannot find reliable information; test representative questions; provide feedback and escalation options; monitor low-rated responses.	AI Lead + QA	**High**
R2	Sensitive customer data is exposed or logged unnecessarily	3	5	15 — High	Collect only necessary data; restrict access by role; avoid storing secrets in prompts/logs; define retention rules; protect transport and stored data; review logs for personal data.	Backend Lead + Security Owner	Medium–High
R3	External LLM or cloud service is unavailable or slow	3	4	12 — High	Set timeouts and controlled retries; show a helpful fallback message; allow ticket creation or human handoff; monitor service errors and response time.	AI/Cloud Developer	**Medium**
R4	AI tool performs an incorrect or unauthorized action	3	5	15 — High	Allowlist available tools; validate every parameter on the server; enforce user permissions independently of the model; require human approval for sensitive actions such as refunds; log actions.	Backend Lead	**Medium**
R5	Knowledge base contains outdated or conflicting articles	3	4	12 — High	Assign article owners; record review dates; provide an update workflow; test retrieval against known questions; show source references where practical.	Knowledge Base Owner + Support Lead	**Medium**
R6	Project scope is too large for the available time	4	4	16 — High	Define a minimum viable product; prioritize core chat, knowledge search, ticket escalation, and basic dashboard; move optional features such as A/B testing and advanced learning to future work.	Project Manager + Team	**Medium**
R7	Integration problems between ASP.NET, AI SDK, search, Redis, and SignalR	4	4	16 — High	Build a small end-to-end prototype early; define interfaces between layers; pin compatible package versions; test integrations separately; keep a documented fallback for nonessential services.	Technical Lead	**Medium**
					Set provider		

R8	API usage costs exceed the project budget	3	4	12 — High	spending limits and alerts where available; cap input/output length; avoid unnecessary repeated calls; cache suitable common responses; monitor usage during testing.	AI/Cloud Developer + Product Owner	Low–Medium
R9	Unauthorized users access agent or admin features	3	5	15 — High	Use ASP.NET Identity roles and authorization policies; apply least privilege; validate permissions on the server; test access-denied cases; protect admin configuration and API credentials.	Backend Lead + QA	**Medium**
R10	Escalation fails or the human queue is not monitored	3	4	12 — High	Define clear escalation triggers; create a ticket if live handoff is unavailable; display ticket status; notify assigned agents; test the full AI-to-human flow.	Support Workflow Owner + Backend Lead	**Medium**
R11	Conversation history or AI-generated summaries lose important context	3	3	9 — Medium	Persist messages reliably; define what a summary must include; let agents inspect the original conversation; test long and multi-turn chats; avoid treating summaries as the sole source of truth.	AI Lead + QA	Low–Medium
R12	Real-time chat updates are lost or duplicated	3	3	9 — Medium	Use stable message IDs and server-side persistence; handle reconnects; avoid relying only on SignalR delivery; test disconnection and retry scenarios.	Backend Lead	Low–Medium
R13	Deployment configuration or secrets are incorrect	3	4	12 — High	Separate development and production settings; store secrets in environment configuration or a managed secret store; verify deployment settings; run smoke tests after release.	DevOps / Deployment Owner	Low–Medium
R14	Analytics show misleading or incomplete metrics	3	3	9 — Medium	Define each metric precisely; calculate metrics from stored events; handle missing feedback explicitly; compare dashboard totals with sample records.	Data/Backend Developer + QA	**Low**
R15	Team members have unclear ownership or duplicated work	3	3	9 — Medium	Assign one owner per feature; document interfaces and deliverables; use a shared task board; review dependencies weekly; integrate work continuously.	Project Manager + Team	**Low**

**Residual risk is a preliminary estimate after the proposed controls; it should be confirmed by the team.*

5. Top Risks Requiring Early Attention

Priority Area	Why It Matters	Immediate Action
1. AI answer quality	The product's main value depends on trustworthy support answers. A wrong answer can frustrate customers or lead them to take an incorrect action.	Test common and edge-case questions; use retrieval from approved articles; provide a clear human escalation path.
2. Privacy and access control	The system stores conversations and may process personal information. A permissions or logging mistake could expose data.	Design role permissions and data handling early; never rely on the AI model to enforce authorization.
3. Tool/action safety	The AI may request an action such as checking an order or initiating a refund. The server must decide whether the action is valid and permitted.	Validate inputs and permissions in C# services; require approval for high-impact actions; maintain an audit trail.
4. Scope and integration complexity	The proposal includes many technologies and advanced capabilities. Attempting all of them at once creates schedule and integration risks.	Deliver a small end-to-end MVP first, then add features in priority order.

6. Recommended Minimum Viable Product (MVP)

To reduce schedule and integration risk, the team should first aim to demonstrate one complete, reliable workflow rather than every advanced feature in the proposal.

Step	MVP Capability
1	Customer sends a message and receives an AI response.
2	The AI searches a small, curated FAQ/knowledge base before answering support questions.
3	Conversation messages are saved in the database.
4	The system creates a support ticket and transfers the case when the answer is uncertain or the customer requests a person.
5	A support agent can view the ticket and conversation summary.
6	Basic authentication, role-based access, error handling, and audit logging are in place.

Features such as advanced A/B testing, automatic knowledge-base improvement, sophisticated learning loops, and extensive analytics can be considered after the core workflow is stable.

7. Risk Monitoring and Review

When	Review Activity
Weekly	Review open risks, changes in likelihood/impact, blocked tasks, and mitigation owners.
At each milestone	Re-score the highest risks and verify that planned controls have been implemented and tested.
Before deployment	Check authorization, sensitive-data handling, API-key storage, error/fallback behavior, and the escalation workflow.
After a demo or user test	Review inaccurate answers, failed searches, abandoned chats, escalation outcomes, and user feedback.

8. Conclusion

The main risks for SmartSupport AI are unreliable AI answers, privacy and authorization failures, unsafe tool execution, dependency on external services, and the complexity of integrating multiple technologies within the project timeline. These risks can be reduced by keeping the initial scope focused, validating every action on the server, grounding answers in approved knowledge, preserving a human handoff, and testing the complete workflow early.

Stakeholder engagement is equally important: customers need understandable and dependable help, agents need a useful escalation workflow, administrators need secure controls, and the project team needs clear ownership and achievable milestones. The risk register should be treated as a living document and updated throughout development.